Unit 9: CNN Model Activity

Introduction

The brief asks me to read Wall (2019), run the supplied convolutional neural network (CNN) on CIFAR-10 (Krizhevsky, 2009), review its sections, and test images 1 to 15 instead of 16.

Running the model

Its two convolution layers pick out small local patterns and combine them into larger ones, and each pooling layer shrinks the image to cut memory use and parameters (Géron, 2025). The model reached a test accuracy of 0.64 and predicted 11 of images 1 to 15 correctly, including image 7, a frog, chosen beforehand. Recall ran from 0.52 for birds to 0.77 for frogs. A high softmax score did not mean a right answer: 330 of the 3,609 errors scored 0.9 or more, such as image 15, a ship called a frog. The supplied early stopping is not set to restore the best weights, so the model kept those from epoch 7, two epochs after the lowest validation loss.

What the data can show

CIFAR-10 holds 32 × 32 pictures of ten object classes, with no person class or demographic labels (Krizhevsky, 2009), so it says nothing about faces. That would need images of the people, cameras and conditions it will meet, labelled by group.

Ethical, social and legal implications

Wall (2019) reports facial recognition errors that differ by skin tone and gender. For the person wrongly matched, an error can mean a wrongful stop. The published figures show how easily a single number misleads, because each answers a

different question. In the Met's 2016–19 trials, only eight of 42 matches were verifiably correct (Fussey and Murray, 2019): that measures how often an alert is right. The Croydon pilot's one false alert among more than 470,000 passers-by (Metropolitan Police, 2026) measures how rarely a passer-by is wrongly flagged. Neither shows who bears the errors. In 2024–25, eight of the ten false alerts involved Black people (Metropolitan Police, 2025), but ten cases, with no count of people scanned by group, cannot give a rate by group. My run cannot test demographic bias, since CIFAR-10 contains no people, but it shows the same trap at object level: one accuracy figure hid uneven recall and confident errors.

Conclusion

In my governance work, I look at whether a review step lets the reviewer challenge an output. I would now ask for error rates by group before sign-off, not one score, and for a view on which uses should proceed at all, the question Wall's (2019) Salesforce example raises. For police use, I would also ask how the deployment shows that its sensitive processing is strictly necessary and meets a Schedule 8 condition (Information Commissioner's Office, 2023).

Evidence from my run

Tables 1 to 7 and Figure 1 come from my Colab run of 11 November 2026 at 14:27 UTC, in which all 56 code cells completed without errors. The model and its settings are as supplied.

Training and test

Table 1: Training and test results. The supplied early stopping watches the validation loss with a patience of 2 and keeps the last epoch's weights.

Measure	Result
Epochs run	7 of a maximum of 25
Lowest validation loss	Epoch 5: 1.0997 (validation accuracy 0.631)
Final weights	Epoch 7: validation loss 1.1111 (validation accuracy 0.646)
Final training accuracy	0.725
Test accuracy, 10,000 images	0.639 (test loss 1.133); the supplied file's saved run gave 0.670

Sections of the model

Table 2: The supplied architecture, from my run's model.summary() output, with the supplied training settings.

Section	Output shape	Parameters
Convolution, 32 filters of 4 × 4, ReLU	29 × 29 × 32	1,568
Max pooling, 2 × 2	14 × 14 × 32	0
Convolution, 32 filters of 4 × 4, ReLU	11 × 11 × 32	16,416
Max pooling, 2 × 2	5 × 5 × 32	0
Flatten	800	0
Dense, 256 units, ReLU	256	205,056
Dense, 10 units, softmax	10	2,570
Training	Adam, categorical cross-entropy, up to 25 epochs, early stopping on validation loss with patience 2	225,610 in total

Errors by class

Table 3: Precision and recall by class on the 10,000 test images (1,000 per class), from the lowest recall to the highest.

Class	Precision	Recall
bird	0.533	0.522
deer	0.614	0.530
cat	0.416	0.546
dog	0.515	0.565
horse	0.723	0.664
airplane	0.754	0.676
ship	0.805	0.698
truck	0.712	0.708
automobile	0.790	0.712
frog	0.658	0.770

Table 4: The five most common mistakes, out of 1,000 test images per true class.

True class	Predicted class	Count
dog	cat	233
cat	dog	177
automobile	truck	136
bird	cat	107
bird	frog	102

Changing the test image

Table 5: Test images 1 to 16. The brief asks for one image from 1 to 15; image 7 was my choice, made before the run, and 16 is the supplied example. Confidence is the model's highest softmax score.

Image	True class	Predicted	Confidence	Correct
1	ship	automobile	0.497	No
2	ship	ship	0.710	Yes
3	airplane	airplane	0.991	Yes
4	frog	deer	0.822	No
5	frog	frog	0.570	Yes
6	automobile	automobile	0.796	Yes
7	frog	frog	0.951	Yes
8	cat	cat	0.526	Yes
9	automobile	automobile	0.930	Yes
10	airplane	ship	0.379	No
11	truck	truck	0.944	Yes
12	dog	dog	0.780	Yes
13	horse	horse	0.963	Yes
14	truck	truck	0.858	Yes
15	ship	frog	0.953	No
16	dog	dog	0.933	Yes



Figure 1: Test images 1 to 16 with the predicted class, its confidence and the true class; errors in red. Pictures and values from my run; the labels were redrawn because the notebook's own grid overlapped them.

Confident errors

Table 6: How confident the model was when it was wrong, on the 10,000 test images.

Measure	Result
Wrong predictions	3,609
Wrong with confidence of 0.5 or more	2,361
Wrong with confidence of 0.8 or more	673
Wrong with confidence of 0.9 or more	330
Average confidence when right / wrong	0.805 / 0.601

Changes to the supplied notebook

Table 7: The changes, also listed in the notebook's opening cell.

Change	Reason
array_to_img imported from keras.utils	It moved from keras.preprocessing.image in current Keras
model.predict_classes replaced by np.argmax over model.predict	predict_classes was removed from Keras
Cells run in order	The supplied file had run the prediction cell out of order and showed a NameError
One comment corrected	The first training image is a frog (label 6), not an automobile

References

Fussey, P. and Murray, D. (2019) *Independent report on the London Metropolitan Police Service's trial of live facial recognition technology*. Colchester: University of Essex Human Rights Centre. Available at: https://repository.essex.ac.uk/24946/ (Accessed: 12 November 2026).

Géron, A. (2025) *Hands-On Machine Learning with Scikit-Learn and PyTorch*. Santa Rosa: O'Reilly.

Information Commissioner's Office (2023) 'Conditions for sensitive processing', *Guide to law enforcement processing*. Available at: https://ico.org.uk/for-organisations/law-enforcement/guide-to-le-processing/conditions-for-sensitive-processing/ (Accessed: 12 November 2026).

Krizhevsky, A. (2009) *Learning multiple layers of features from tiny images*. Technical report. Toronto: University of Toronto. Available at: https://cave.cs.toronto.edu/kriz/cifar.html (Accessed: 11 November 2026).

Metropolitan Police (2025) *Live facial recognition annual report September 2025*. London: Metropolitan Police Service. Available at: https://www.met.police.uk/SysSiteAssets/media/downloads/force-content/met/advice/lfr/other-lfr-documents/live-facial-recognition-annual-report-2025.pdf (Accessed: 11 November 2026).

Metropolitan Police (2026) *Met makes one arrest every 35 minutes during live facial recognition pilot*, 13 May. Available at: https://news.met.police.uk/news/met-makes-one-arrest-every-35-minutes-during-live-facial-recognition-pilot-509256 (Accessed: 12 November 2026).

Wall, M. (2019) 'Biased and wrong? Facial recognition tech in the dock', *BBC News*, 8 July. Available at: https://www.bbc.com/news/business-48842750 (Accessed: 20 October 2026).

Supplied materials: University of Essex Online Unit 9 e-Portfolio activity brief and notebook, Convolutional Neural Networks (CNN) - Object Recognition.ipynb.

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